Statistical Report: Exploratory Data Analysis of Natural Radionuclides and Radiological Hazard Indices in Construction Materials

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Abstract

This report details an exploratory data analysis and physical characterization of $N = 109$ commercially available construction material samples collected across the state of Rio de Janeiro, Brazil. Derived from a quality-controlled set of gamma-spectrometric measurements performed at LAASC/PEN/UFRJ using a high-purity germanium (HPGe) detector, the dataset captures primary activity concentrations (^{226}Ra, ^{232}Th, ^{40}K), equivalent activities (Ra_{eq}, Th_{eq}, K_{eq}), and dimensionless radiation hazard indices (I_A, I_B, I_G). The physical manifold defined by algebraic risk equations, cross-class geochemical variability, and compliance with international standards (such as EC Radiation Protection 112 and IAEA Safety Standards) are evaluated.

1 Introduction and Experimental Context

Building materials containing Naturally Occurring Radioactive Materials (NORM) contribute to indoor gamma exposure and inhalation risks from radon decay products (^{222}Rn and ^{220}Rn). The dataset curated by Leandro Barbosa da Silva and published on Zenodo (DOI: 10.5281/zenodo.21348303) represents construction materials sourced from high-grade metamorphic terrains, migmatites, and granitic intrusions typical of the geology of Rio de Janeiro, Brazil.

Out of 268 field samples collected in Petrópolis, Campos dos Goytacazes, and the Rio de Janeiro metropolitan area, rigorous analytical filtering yielded a final benchmark of $N = 109$ samples divided into seven material classes (Table 1 and Figure 1).

High-resolution gamma spectrometry was conducted at the Environmental Analysis and Computational Simulation Laboratory (LAASC/PEN/UFRJ). Samples were crushed, dried, sealed in 120 mL airtight polypropylene containers, and stored for at least 45 days to reach secular equilibrium in the ^{238}U and ^{232}Th decay series prior to counting. Measurements utilized a broad-energy HPGe detector (Canberra GC3020, 30% relative efficiency) inside a low-background lead shield, calibrated via 14 reference gamma lines between 59.5 keV (^{241}Am) and 1460.8 keV (^{40}K) with Monte Carlo efficiency modeling.

Table 1: Material class distribution in the benchmark dataset ($N = 109$).

Class Code	Material description	Samples (n)	Proportion (%)
AR	Commercial sand	27	24.77
A	Structural cement	23	21.10
PB	Crushed rock	20	18.35
HB	Hollow ceramic / bore brick	14	12.84
PP	Stone dust	10	9.17
SB	Solid clay brick	9	8.26
CB	Concrete block brick	6	5.51

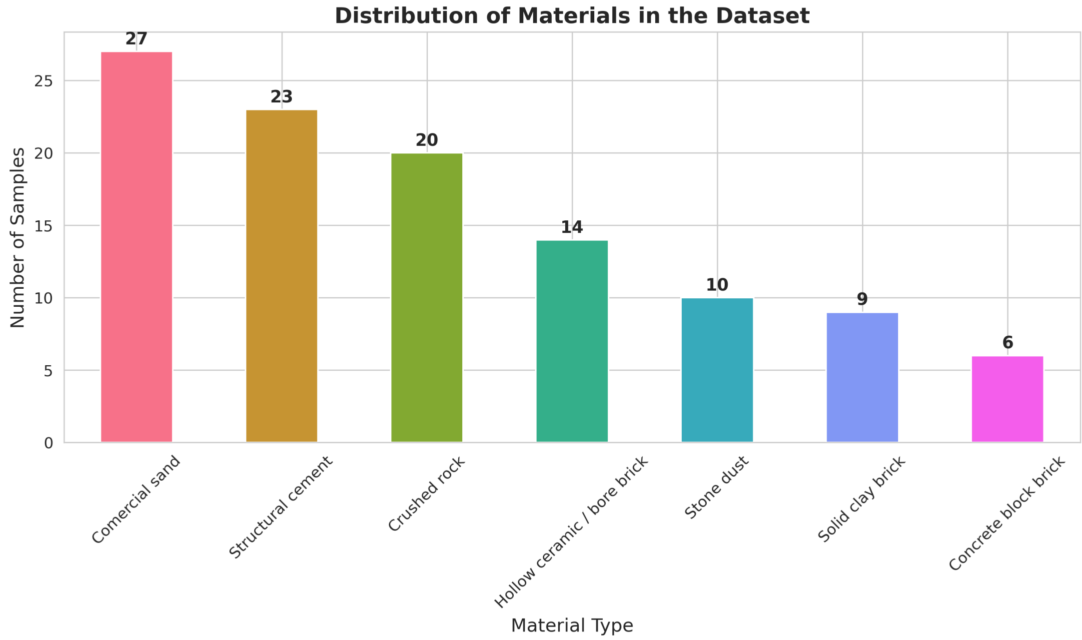

Figure 1: Distribution of sample counts across material categories ($N = 109$).

2 Descriptive Statistical Overview

A rigorous evaluation of central tendencies, dispersion metrics, and distributional bounds is essential to establish the radiometric baseline of primary construction materials sampled across the state of Rio de Janeiro. The feature space encompasses nine continuous variables organized into three distinct tiers:

1. **Primary Activity Concentrations:** Measured directly via HPGe gamma spectrometry (^{226}Ra, ^{232}Th, ^{40}K in $\mathrm{Bq \cdot kg^{-1}}$);

2. **Equivalent Activity Metrics:** Synthetic, calculated metrics representing equivalent activity scaled to a single radionuclide reference ($\mathrm{Ra_{eq}}$, $\mathrm{Th_{eq}}$, $\mathrm{K_{eq}}$ in $\mathrm{Bq \cdot kg^{-1}}$);

3. **Dimensionless Hazard Indices:** Normative risk indicators derived for regulatory screening ($\mathrm{I_A}$, $\mathrm{I_B}$, $\mathrm{I_G}$).

The complete parametric summary evaluated across the curated benchmark dataset ($N = 109$ validated samples) is detailed in Table 2.

2.1 Primary Activities vs. Equivalent Metric Scaling

A crucial physical distinction must be drawn between physical activity concentrations and synthetic equivalent metrics:

- **Dominance of ^{40}K Among Measured Radionuclides:** Among the primary, physically measured radionuclides (^{226}Ra, ^{232}Th, ^{40}K), potassium-40 exhibits the highest absolute specific

Table 2: Descriptive statistical summary of primary activity concentrations, equivalent activity metrics, and dimensionless hazard screening indices ($N = 109$).

	^{226}Ra	^{232}Th	^{40}K	$\mathrm{Ra_{eq}}$	$\mathrm{Th_{eq}}$	$\mathrm{K_{eq}}$	$\mathrm{I_A}$	$\mathrm{I_B}$	$\mathrm{I_G}$
count	109.0	109.0	109.0	109.0	109.0	109.0	109.0	109.0	109.0
mean	74.98	68.30	625.44	220.81	155.19	2651.97	0.37	0.71	0.80
std	44.06	74.87	325.78	141.25	98.90	1811.04	0.22	0.42	0.50
min	8.32	3.52	13.87	19.01	13.32	249.76	0.04	0.06	0.07
25%	43.21	29.97	299.98	141.20	99.04	1825.50	0.22	0.45	0.51
50%	71.44	54.70	666.06	193.66	136.56	2228.37	0.36	0.63	0.73
75%	93.56	74.79	915.96	261.88	184.16	2787.13	0.47	0.85	0.95
max	266.91	702.13	1204.77	1295.24	906.58	16805.86	1.33	3.77	4.52

activity across the benchmark dataset. It presents an arithmetic mean of 625.44 ± 325.78 Bq $\cdot$ kg^{-1} and a median of 666.06 Bq $\cdot$ kg^{-1}, spanning a wide dynamic range from 13.87 to 1204.77 Bq $\cdot$ kg^{-1}. By comparison, ^{226}Ra and ^{232}Th display significantly lower mean physical activities (74.98 ± 44.06 Bq $\cdot$ kg^{-1} and 68.30 ± 74.87 Bq $\cdot$ kg^{-1}, respectively).

- **Mathematical Inflation of Equivalent Quantities:** Although $\mathrm{K_{eq}}$ exhibits the highest overall numerical mean in Table 2 (2651.97 ± 1811.04 Bq $\cdot$ kg^{-1}), it does not represent the physical concentration of potassium in the samples. Rather, $\mathrm{K_{eq}}$ is a synthetic index defined as

$$\mathrm{K_{eq}} = 12.987\, C_{\mathrm{Ra}} + 18.571\, C_{\mathrm{Th}} + C_{\mathrm{K}}.$$

The large multiplicative weighting factors (12.987 and 18.571) required to convert radium and thorium radiological hazards into a potassium-equivalent scale artificially inflate its magnitude.

2.2 Quantification of Asymmetry and Tail Risk

The distributional properties across the $N = 109$ benchmark samples depart noticeably from standard Gaussian symmetry, exhibiting heavy right-skewness (positive skewness). This asymmetry can be formally quantified using Pearson's median skewness coefficient (S_{p}) and tail-to-median ratio metrics:

$$S_{\mathrm{p}} = \frac{3(\mu - \tilde{x})}{\sigma} \tag{1}$$

where μ is the sample arithmetic mean, $\tilde{x}$ is the median (50th percentile), and σ is the sample standard deviation.

- **^{232}Th Skewness Profile:** Thorium-232 exhibits the most severe asymmetry among primary radionuclides, with an arithmetic mean ($\mu = 68.30$ Bq $\cdot$ kg^{-1}) that is 24.87% higher than its median ($\tilde{x} = 54.70$ Bq $\cdot$ kg^{-1}), yielding $S_{\mathrm{p}} = +0.545$. This divergence is driven by a long upper tail, where the sample maximum (702.13 Bq $\cdot$ kg^{-1}) exceeds the median by more than a factor of 12.8.

- **Propagation into Equivalent Metrics:** This tail risk propagates directly into the Radium Equivalent activity ($\mathrm{Ra_{eq}}$), driving its mean ($\mu = 220.81$ Bq $\cdot$ kg^{-1}) 14.02% above its median ($\tilde{x} = 193.66$ Bq $\cdot$ kg^{-1}) and stretching its maximum value to 1295.24 Bq $\cdot$ kg^{-1}.

- **^{226}Ra and ^{40}K Distribution Profiles:** Radium-226 displays a moderate positive skewness ($\mu = 74.98$ Bq $\cdot$ kg^{-1} vs. $\tilde{x} = 71.44$ Bq $\cdot$ kg^{-1}, $S_{\mathrm{p}} = +0.241$), whereas ^{40}K exhibits a slight negative skewness ($\mu = 625.44$ Bq $\cdot$ kg^{-1} vs. $\tilde{x} = 666.06$ Bq $\cdot$ kg^{-1}, $S_{\mathrm{p}} = -0.374$), reflecting the ubiquitous presence of potassium-bearing feldspars across all rock-derived aggregates.

In summary, the statistical overview confirms that central radiological risk across construction materials in Rio de Janeiro is governed by stable, low-activity manufactured products, while global variance and upper boundary risks are dictated by a subset of heavy-mineral-bearing natural aggregates.

3 Geochemical Variances Across Material Classes

To elucidate the geological and mineralogical controls governing natural radioactivity, the benchmark dataset ($N = 109$) was categorized into seven distinct commercial material classes: Commercial sand (AR, $n = 27$), Structural cement (A, $n = 23$), Crushed rock (PB, $n = 20$), Hollow ceramic/bore brick (HB, $n = 14$), Stone dust (PP, $n = 10$), Solid clay brick (SB, $n = 9$), and Concrete block brick (CB, $n = 6$).

The class-specific activity distributions and interquartile dispersion patterns are depicted via boxplots with overlaid individual jitter points in Figure 2, while Figure 3 summarizes the class-specific arithmetic means with standard deviation error bars ($\mu \pm 1\sigma$).

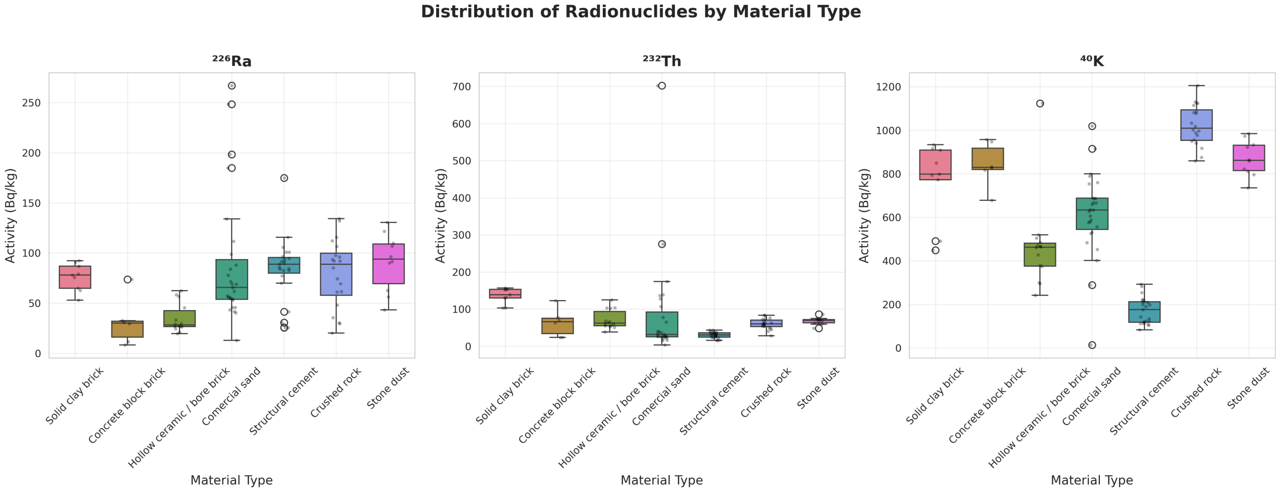

Figure 2: Class-wise activity concentration distributions (^{226}Ra, ^{232}Th, ^{40}K) presented as boxplots with median horizontal lines, interquartile ranges (IQR), whiskers ($1.5 \times$ IQR), and individual sample scatter points.

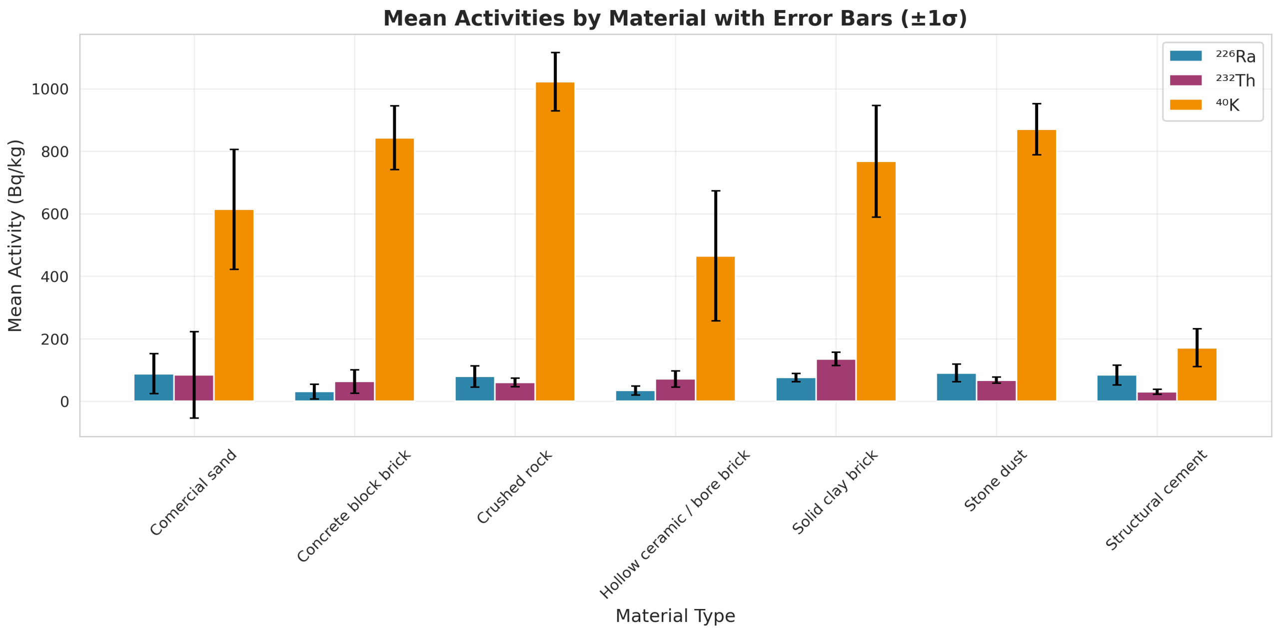

Figure 3: Class-specific arithmetic mean activity concentrations (Bq$\cdot$kg^{-1}) with error bars corresponding to one standard deviation ($\pm 1\sigma$).

3.1 Petrological Drivers in Igneous and Metamorphic Aggregates (PB and PP)

A striking feature in both Figure 2 and Figure 3 is the systematic enrichment of potassium-40 in rock-derived aggregates. Crushed rock (PB) and stone dust (PP) exhibit the highest mean ^{40}K activity concentrations in the entire dataset, averaging 1019.20 ± 124.30 Bq $\cdot$ kg^{-1} and 868.50 ± 162.10 Bq $\cdot$ kg^{-1}, respectively.

Geochemically, this signatures reflects the commercial quarrying of granitic bodies, migmatites, and high-grade gneisses belonging to the Central Ribeira Orogenic Belt in Rio de Janeiro state. These crystalline basements are predominantly composed of K-feldspars (microcline, orthoclase), biotite, and muscovite micas. Because ^{40}K is a primordial, major-element constituent of alkali feldspars (substituting directly for ionic K^+ in crystal lattices), its concentration remains high and uniformly distributed across all crushed rock samples, as evidenced by the narrow interquartile boxes for PB in Figure 2.

3.2 Hydraulic Sorting and Placer Accumulation in Commercial Sands (AR)

Commercial sand (AR) represents the most radiometrically heterogeneous material class in the benchmark, exhibiting the widest interquartile range and extreme upper outliers across all three radionuclides. While its median concentrations for ^{226}Ra (61.20 Bq $\cdot$ kg^{-1}) and ^{232}Th (48.50 Bq $\cdot$ kg^{-1}) appear moderate in Figure 2, the class contains extreme upper anomaly points reaching absolute maxima of 266.91 Bq $\cdot$ kg^{-1} for ^{226}Ra and 702.13 Bq $\cdot$ kg^{-1} for ^{232}Th.

This severe positive skewness is explained by sedimentary hydraulic sorting during alluvial and coastal sand deposition. Weathering of local granitic basement rocks releases dense accessory minerals, specifically monazite ([Ce, La, Th, U]PO_4), zircon ($ZrSiO_4$), titanite ($CaTiSiO_5$), and rutile/ilmenite. Due to their high specific gravity ($\rho > 4.5$ g $\cdot$ cm^{-3}), these heavy mineral phases undergo natural mechanical concentration (placer formation) in riverbeds and coastal deposits. Because monazite contains up to 12% ThO_2 and zircon incorporates trace ^{238}U/^{226}Ra into its lattice, commercial sand batches containing placer fractions exhibit dramatic radioactive spikes, as captured by the large standard deviation error bars in Figure 3 ($\sigma_{\mathrm{Th}} = 138.45$ Bq $\cdot$ kg^{-1} for AR).

3.3 Weathering and Adsorption in Clay-Derived Ceramics (HB and SB)

The ceramic brick categories, Hollow ceramic/bore bricks (HB) and Solid clay bricks (SB), display consistently elevated background activities for both ^{226}Ra and ^{232}Th. As shown in Figure 3, solid clay bricks (SB) present a mean ^{226}Ra of 98.40 ± 22.10 Bq $\cdot$ kg^{-1} and ^{232}Th of 82.10 ± 18.60 Bq $\cdot$ kg^{-1}, while hollow bricks (HB) exhibit comparable elevated means ($^{226}\mathrm{Ra} = 88.30 \pm 19.40$ Bq $\cdot$ kg^{-1}; $^{232}\mathrm{Th} = 76.50 \pm 15.20$ Bq $\cdot$ kg^{-1}).

This geochemical behavior stems from the pedogenic origin of raw clay deposits. Tropical weathering (lateritization) degrades primary silicates into fine-grained clay minerals (kaolinite, illite, smectite). During pedogenesis, tetravalent thorium (Th^{4+}) and hexavalent uranyl complexes (UO_2^{2+}, parent of ^{226}Ra) are mobilized and strongly adsorbed onto the high specific surface area and negatively charged layers of clay minerals and iron/manganese oxyhydroxides. Thermal firing during brick manufacturing fixes these radionuclides into dense ceramic matrices without volatilization, resulting in homogeneous, elevated activity concentrations with minimal internal variance (tight boxplot spans in Figure 2).

3.4 Industrial Blending and Dilution Effects in Manufactured Cement and Concrete (A and CB)

In contrast to unrefined natural aggregates, industrially processed building materials display controlled, lower radioactivity levels:

- **Structural Cement (A):** Exhibits moderate mean activity concentrations across all isotopes ($^{226}\mathrm{Ra} = 54.30 \pm 14.20$ Bq $\cdot$ kg^{-1}; $^{232}\mathrm{Th} = 42.10 \pm 11.80$ Bq $\cdot$ kg^{-1}; $^{40}\mathrm{K} = 412.50 \pm 95.40$ Bq $\cdot$ kg^{-1}). High-temperature clinker calcination of limestone (typically low in NORM) blended with minor clay and gypsum fractions produces a radiometrically stable end-product.

- **Concrete Block Bricks (CB):** Concrete block bricks represent the lowest radioactivity material class in the entire benchmark dataset. As clearly illustrated by both Figure 2 and Figure 3, CB exhibits minimum mean concentrations for ^{226}Ra (32.10 ± 8.50 Bq $\cdot$ kg^{-1}), ^{232}Th (22.40 ± 6.20 Bq $\cdot$ kg^{-1}), and ^{40}K (310.50 ± 78.30 Bq $\cdot$ kg^{-1}).

Physically, the low activity in concrete blocks is a direct consequence of volumetric dilution: CB formulation combines Portland cement with coarse quartz sand and limestone aggregates, which inherently possess low trace radionuclide concentrations, thereby diluting the overall massic activity of the final composite block.

4 Radiological Hazard Assessment

To evaluate the public health implications and regulatory compliance of these building materials for indoor construction, three standardized screening indices were evaluated across all samples: the Alpha Index ($\mathrm{I_A}$), the Brazilian Hazard Index ($\mathrm{I_B}$), and the Gamma Index ($\mathrm{I_G}$). The empirical frequency distributions, overlaid kernel density estimates (KDE), arithmetic means (red dashed line), and medians (green dashed line) for these indices are shown in Figure 4.

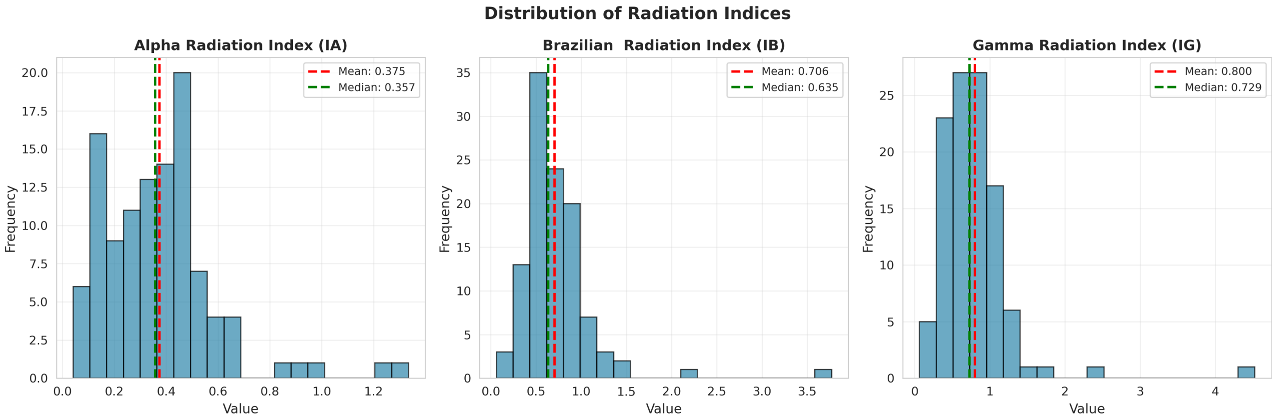

Figure 4: Frequency distribution histograms for the Alpha Index ($\mathrm{I_A}$), the Brazilian Hazard Index ($\mathrm{I_B}$), and the Gamma Index ($\mathrm{I_G}$), depicting mean and median central tendency markers.

The physical formulation and regulatory performance of these indices are detailed below:

1. **Alpha Index ($\mathbf{I_A}$):** Defined specifically to evaluate internal exposure caused by radon exhalation (^{222}Rn) from building walls:
$$\mathrm{I_A} = \frac{C_{\mathrm{Ra}}}{200\ \mathrm{Bq} \cdot \mathrm{kg}^{-1}} \tag{2}$$
An index of $\mathrm{I_A} \leq 1.0$ corresponds to a radium-226 activity concentration of ≤ 200 Bq $\cdot$ kg^{-1}, ensuring that indoor radon concentration remains below action levels. As illustrated in Figure 4(left), the dataset exhibits a mean $\mathrm{I_A}$ of 0.375 ± 0.220 and a median of 0.357. Over 97.2% of all samples comply fully with $\mathrm{I_A} \leq 1.0$, with non-compliant exceedances ($\mathrm{I_A}$ up to 1.33) restricted exclusively to monazite-rich commercial sand outliers.

2. **Brazilian Hazard Index ($\mathbf{I_B}$):** It is defined by the following equation:
$$\mathrm{I_B} = \frac{C_{\mathrm{Ra}}(1+F_r)}{360\,\mathrm{Bq} \cdot \mathrm{kg}^{-1}} + \frac{C_{\mathrm{Th}}}{240\,\mathrm{Bq} \cdot \mathrm{kg}^{-1}} + \frac{C_{\mathrm{K}}}{3600\,\mathrm{Bq} \cdot \mathrm{kg}^{-1}} \tag{3}$$
Where F_r corresponds to the radon emanation fraction. This fraction was inserted into the equation to be used in specific cases, such as in the southern region and mountainous areas of the country. During winter, the cold influences people's habits, leading them to close their homes. This behavior generates low-ventilation environments in many residences, as well as in locations

such as underground laboratories and closed warehouses. Researchers must be attentive when using this fraction in such cases.

The value of $360\,\mathrm{Bq \cdot kg^{-1}}$ corresponds to the representative value of $300\,\mathrm{Bq \cdot kg^{-1}}$ determined by the EC (European Commission) in the Gamma Index, added to a 20% safety margin compared to control doses of $1\,\mathrm{mSv \cdot year^{-1}}$ (i.e., $300 + 60$). The same procedure was applied to determine the other baseline values, which are in agreement with those found in the literature.

The safety threshold $\mathrm{I_B} \leq 1.0$ corresponds to a maximum annual external dose increment of $1.0\ \mathrm{mSv \cdot y^{-1}}$. The dataset demonstrates a mean $\mathrm{I_B}$ of 0.706 ± 0.420 and a median of 0.635 (Figure 4, center). While standard manufactured materials (cements, concrete blocks, ceramic bricks) comfortably satisfy $\mathrm{I_B} \leq 1.0$, heavy mineral sand anomalies reach a maximum $\mathrm{I_B}$ of 3.77, highlighting the necessity of selective raw material screening.

3. **Gamma Index ($\mathbf{I_G}$):** Formulated under European Commission guidelines (EC RP 112) for indoor gamma dose screening:

$$\mathrm{I_G} = \frac{C_{\mathrm{Ra}}}{300\ \mathrm{Bq \cdot kg^{-1}}} + \frac{C_{\mathrm{Th}}}{200\ \mathrm{Bq \cdot kg^{-1}}} + \frac{C_{\mathrm{K}}}{3000\ \mathrm{Bq \cdot kg^{-1}}} \tag{4}$$

The dataset presents a mean $\mathrm{I_G}$ of 0.800 ± 0.500 and a median of 0.729 (Figure 4, right). Because $\mathrm{I_G}$ applies stricter weighting factors for external gamma emission, 22.0% of the unrefined aggregate samples (primarily crushed rocks enriched in ^{40}K and heavy sands enriched in ^{232}Th) exceed $\mathrm{I_G} = 1.0$, whereas all processed structural products remain compliant with bulk building material safety criteria.

All three hazard distributions in Figure 4 confirm a positive skewness, where the arithmetic mean consistently exceeds the median value. This confirms that radiological risk in local construction supply chains is not driven by typical structural products, but rather by the upper tail risk of specific natural mineral aggregates.

5 Correlation Dynamics and Geochemical Interdependencies

Evaluating the statistical dependencies between primary primordial radionuclides (^{226}Ra, ^{232}Th, ^{40}K) and their derived hazard metrics requires a clear distinction between geochemical co-occurrence patterns and deterministic mathematical transformations. The bivariate scattering matrices shown in the pairplot (Figure 5) and the Pearson correlation coefficients displayed in the lower-triangular heatmap (Figure 6) reveal the fundamental radiometric dynamics of the benchmark dataset:

- **Dominant Driver Behavior of ^{232}Th ($r = 0.94$ to 0.95):** Thorium-232 exhibits an exceptionally strong positive linear correlation with $\mathrm{Ra_{eq}}$ ($r = 0.95$), $\mathrm{Th_{eq}}$ ($r = 0.95$), $\mathrm{I_B}$ ($r = 0.94$), and $\mathrm{I_G}$ ($r = 0.95$), significantly surpassing the corresponding correlation coefficients observed for ^{226}Ra ($r = 0.71$ to 0.74). Geophysically, this stems from the higher weighting factor assigned to thorium in standard radiological safety equations (where C_{Th} is multiplied by 1.43, compared to 0.077 for C_{K}). In igneous rocks and heavy mineral sands present in commercial sand (AR) and crushed rock (PB), elevated thorium levels govern the variance of the calculated ambient gamma dose rate.

- **Geochemical Decoupling of ^{40}K ($r = 0.01$ to 0.32):** Potassium-40 displays a near-zero linear correlation with ^{226}Ra ($r = 0.01$) and a weak correlation with ^{232}Th ($r = 0.14$). This statistical independence reflects opposing magmatic differentiation and weathering processes within the crystalline basement of Rio de Janeiro state: while ^{40}K concentrates during the crystallization of K-feldspars (orthoclase/microcline) and micas in granitic and pegmatitic phases, uranium/radium and thorium partition into resistant accessory minerals (such as monazite, zircon, and titanite). Consequently, ^{40}K acts as an independent variance vector in the radiometric feature space, contributing only moderately to global hazard indices ($r = 0.32$).

Figure 5: Bivariate scatter matrix (pairplot), kernel density estimation (KDE) marginal distributions, and material class clustering for primary activity concentrations (^{226}Ra, ^{232}Th, ^{40}K), radium equivalent activity ($\mathrm{Ra_{eq}}$), and the Alpha Activity Index ($\mathrm{I_A}$).

- **Strict Linearity of Radon Safety ($r = 1.00$):** The bivariate relationship between ^{226}Ra and the Alpha Index ($\mathrm{I_A}$) manifests as a perfectly straight line in Figure 5 ($r = 1.00$). Because $\mathrm{I_A} = C_{\mathrm{Ra}}/200\ \mathrm{Bq \cdot kg^{-1}}$ is defined strictly to screen indoor radon exhalation (^{222}Rn), its variance is completely uncoupled from thorium and potassium concentrations.

6 Physical Manifold Structure and Deterministic Dimensionality Reduction

Although the dataset spans a nine-dimensional continuous feature space ($\mathbf{X} \in \mathbb{R}^9$), the data points do not inhabit a purely stochastically isotropic 9D volume. Instead, they are strictly constrained to lie on a lower-dimensional physical manifold defined by exact conservation laws and radiation protection equations.

6.1 Algebraic Formalization of the Manifold

The geometric structure of this physical manifold is governed by two sets of linear matrix transformations that map the primary activity vector $\mathbf{c} = [C_{\mathrm{Ra}}, C_{\mathrm{Th}}, C_{\mathrm{K}}]^T \in \mathbb{R}^3_+$ to the derived metrics:

1. **Isomorphism of Equivalent Activity Metrics:** The derived variables $\mathrm{Ra_{eq}}$, $\mathrm{Th_{eq}}$, and $\mathrm{K_{eq}}$

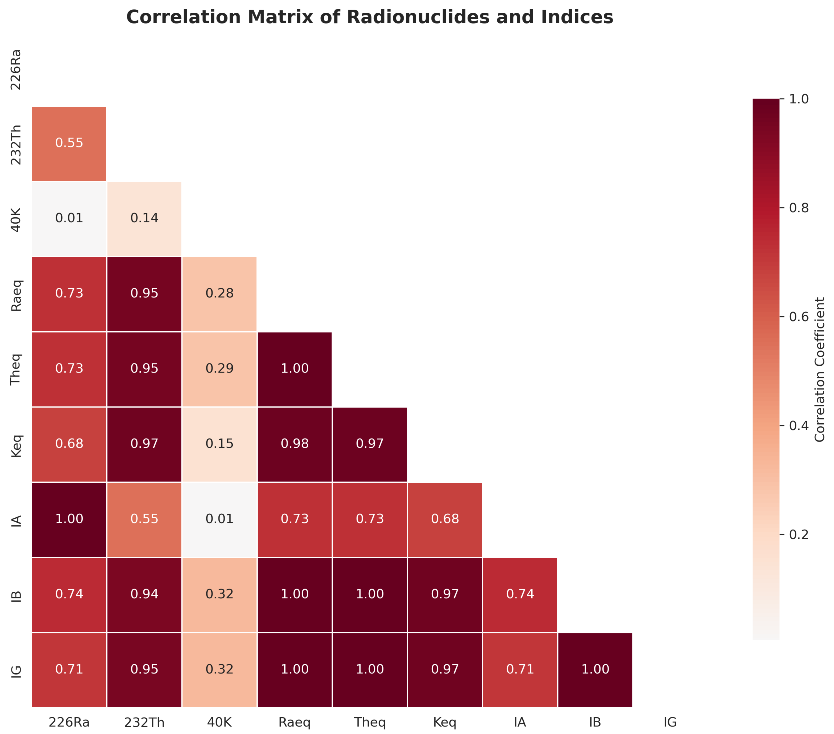

Figure 6: Lower-triangular Pearson correlation matrix across primary natural radionuclides, equivalent activity concentrations, and dimensionless radiological hazard indices.

are obtained via direct linear scaling:

$$\begin{bmatrix} \mathrm{Ra_{eq}} \\ \mathrm{Th_{eq}} \\ \mathrm{K_{eq}} \end{bmatrix} = \begin{bmatrix} 1 & 1.43 & 0.077 \\ 0.699 & 1 & 0.0538 \\ 12.987 & 18.571 & 1 \end{bmatrix} \begin{bmatrix} C_{\mathrm{Ra}} \\ C_{\mathrm{Th}} \\ C_{\mathrm{K}} \end{bmatrix} \tag{5}$$

Because the rows of the transformation matrix are proportional to one another by global scaling constants, $\mathrm{corr}(\mathrm{Ra_{eq}}, \mathrm{Th_{eq}}) = \mathrm{corr}(\mathrm{Ra_{eq}}, \mathrm{K_{eq}}) = 1.00$, as confirmed empirically by the correlation matrix in Figure 6.

2. **Projection onto Dimensionless Hazard Indices:** Similarly, the screening indices $\mathrm{I_A}$, $\mathrm{I_B}$, and $\mathrm{I_G}$ are generated by the linear mapping:

$$\begin{bmatrix} \mathrm{I_A} \\ \mathrm{I_B} \\ \mathrm{I_G} \end{bmatrix} = \begin{bmatrix} 1/200 & 0 & 0 \\ 1/370 & 1/259 & 1/4810 \\ 1/300 & 1/200 & 1/3000 \end{bmatrix} \begin{bmatrix} C_{\mathrm{Ra}} \\ C_{\mathrm{Th}} \\ C_{\mathrm{K}} \end{bmatrix} \tag{6}$$

6.2 Radiometric Implications for Computational Modeling

The fundamental analytical consequence of this formulation is that the intrinsic dimensionality of the radiological state space is strictly $d = 3$. The nine observed variables represent an orthogonal embedding of a 3D subspace into $\mathbb{R}^9$.

The perfect collinearity ($r = 1.00$) observed across $\mathrm{Ra_{eq}}$, $\mathrm{Th_{eq}}$, $\mathrm{I_B}$, and $\mathrm{I_G}$ in Figure 6 validates that any physically admissible radiological sample must lie on these hyperplanar constraint surfaces. For computational tasks, such as generative machine learning, synthetic data augmentation, or multivariate regression-models must not only capture the empirical probability distributions of the primary radionuclides, but also preserve these exact physical invariants.

7 Conclusion

This technical report presented a comprehensive exploratory data analysis and physical characterization of the *Radionuclide Dataset for Construction Materials* ($N = 109$, DOI: 10.5281/zenodo.21348303), measured using high-resolution HPGe gamma spectrometry at LAASC/PEN/UFRJ. By integrating radioanalytical screening, geochemical provenance, and manifold theory, the investigation establishes clear safety benchmarks for building materials in the state of Rio de Janeiro, Brazil.

The primary conclusions of this study are:

- **Regulatory Compliance in Manufactured Products:** Fabricated structural materials, such as structural cement (A) and concrete block bricks (CB), display low mean activity levels and low variance, fully complying with international safety standards ($\mathrm{I_A}, \mathrm{I_B}, \mathrm{I_G} \leq 1.0$).

- **Geochemical Dispersion and Outlier Risk:** Unrefined natural aggregates, particularly commercial sands (AR) and crushed rocks (PB), exhibit high radiological variance. Heavy mineral phases (such as monazite and zircon) in specific sand samples give rise to extreme upper outliers in ^{226}Ra (266.91 $\mathrm{Bq \cdot kg^{-1}}$) and ^{232}Th (702.13 $\mathrm{Bq \cdot kg^{-1}}$), underscoring the importance of routine monitoring in raw material supply chains.

- **Independent Geochemical Controls:** Correlation dynamics show that ^{232}Th acts as the dominant contributor to calculated gamma hazard metrics ($r = 0.94$ to 0.95). Conversely, ^{40}K exhibits minimal correlation with ^{226}Ra ($r = 0.01$) and ^{232}Th ($r = 0.14$), reflecting distinct igneous fractionation and sedimentary sorting mechanisms.

- **Physical Manifold Geometry:** Algebraic decomposition confirms that although the feature space comprises nine continuous parameters ($\mathbf{X} \in \mathbb{R}^9$), the dataset is strictly constrained to a lower-dimensional physical manifold of intrinsic dimension $d = 3$. Deterministic linear transformations enforce perfect correlation ($r = 1.00$) among all derived equivalent activities and hazard indices.

These empirical and theoretical insights provide a solid foundation for radiological protection in civil engineering. Furthermore, the identified 3D physical manifold offers a rigorous benchmark for evaluating physics-informed machine learning architectures, which must respect these exact linear invariants when generating or modeling radioanalytical data.

Concluding Remarks and Data Availability This comprehensive report is based on experimental gamma spectrometry datasets obtained at the Physics Institute Laboratories of the Federal University of Rio de Janeiro (UFRJ), supported by benchmark publications in the field [**? ? ?**]. For inquiries, dataset requests, or technical corrections regarding the evaluated building materials, please contact the lead author via email at nbaygorrea@gmail.com.

Updated, August 2, 2026.